

CM0859 – MT5009 Type Theory

Typed Lambda Calculus (or Proofs as Programs)

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Preliminaries

Textbook

“Lecture Notes on Proofs as Programs” (Pfenning 2023).

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“Lecture Notes on Proofs as Programs” (Pfenning 2023).

Conventions

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Typed Lambda Calculus (or Proofs as Programs)

Introduction

- Propositions (types): $A, B, C, \dots$

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- Lambda terms (programs) : $M, N, \dots$

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- Propositions (types): $A, B, C, \dots$
- Lambda terms (programs) : $M, N, \dots$
- Judgements:
 - (i) A type (A is a type)
 - (ii) $M : A$
(λ -term M is a proof-term for proposition A)
(program M has type A)

Typed Lambda Calculus (or Proofs as Programs)

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- Propositions (types): $A, B, C, \dots$
- Lambda terms (programs) : $M, N, \dots$
- Judgements:
 - (i) A type (A is a type)
 - (ii) $M : A$
 - (λ -term M is a proof-term for proposition A)
 - (program M has type A)
- Proposition-as-types principle

“We intend that if $M : A$ then A true. Conversely, if A true then $M : A$ for some appropriate proof-term M . But we want something more: every deduction of $M : A$ should correspond to a deduction of A true with an identical structure and vice versa.” (p. L4.1)

(continued on next slide)

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Introduction (continuation)

- Form of the inference rules (same than for natural deduction)

$$\frac{J_1 \quad \dots \quad J_n}{J} \text{ rule name}$$

where J (conclusion) and $J_1, \dots, J_n$ (premises) are all judgements.

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where J (conclusion) and $J_1, \dots, J_n$ (premises) are all judgements.

- Types of inference rules: Formation, introduction, elimination and **computation** rules.

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Observation

Other names for the proposition-as-types principle are (or should be):

- the Curry-Howard correspondence,
- the Curry-Howard isomorphism (textbook),
- the Brouwer-Heyting-Kolmogorov-Schönfinkel-Curry-Meredith-Kleene-Feys-Gödel-Läuchli-Kreisel-Tait-Lawvere-Howard-de Bruijn-Scott-Martin Löf-Girard-Reynolds-Stenlund-Constable-Coquand-Huet · · · correspondence (Sørensen and Urzyczyn 2006, p. viii).

Typed Lambda Calculus (or Proofs as Programs)

Definition

The **types** of typed lambda calculus are defined by the following grammar:

Types $\ni \sigma, \tau ::= \mathbf{b}$	(base types)
$\sigma \rightarrow \tau$	(function types)
$\sigma \times \tau$	(product types)

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Definition

The λ -**terms** of our typed lambda calculus are defined by the following grammar:

$M, N ::= x$	(variables)
$\lambda x.M$	(λ -abstractions)
$M N$	(applications)
$\langle M, N \rangle$ $\text{fst } M$ $\text{snd } M$	(pairs and projections)

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Inference rules for conjunction (product types)

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Inference rules for conjunction (product types)

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \wedge B \text{ prop}}$$

$$\frac{A \text{ type} \quad B \text{ type}}{A \times B \text{ type}}$$

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Inference rules for conjunction (product types)

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \wedge B \text{ prop}}$$

$$\frac{A \text{ type} \quad B \text{ type}}{A \times B \text{ type}}$$

- Introduction rule

$$\frac{A \text{ true} \quad B \text{ true}}{A \wedge B \text{ true}} \wedge I$$

$$\frac{M : A \quad N : B}{\langle M, N \rangle : A \wedge B} \wedge I$$

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- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \wedge B \text{ prop}}$$

$$\frac{A \text{ type} \quad B \text{ type}}{A \times B \text{ type}}$$

- Introduction rule

$$\frac{A \text{ true} \quad B \text{ true}}{A \wedge B \text{ true}} \wedge I$$

$$\frac{M : A \quad N : B}{\langle M, N \rangle : A \wedge B} \wedge I$$

- Elimination rules

$$\frac{A \wedge B \text{ true}}{A \text{ true}} \wedge E_1$$

$$\frac{M : A \wedge B}{\text{fst } M : A} \wedge E_1$$

$$\frac{A \wedge B \text{ true}}{B \text{ true}} \wedge E_2$$

$$\frac{M : A \wedge B}{\text{snd } M : B} \wedge E_2$$

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Inference rules for implication (function types)

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Inference rules for implication (function types)

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \supset B \text{ prop}}$$

$$\frac{A \text{ type} \quad B \text{ type}}{A \rightarrow B \text{ type}}$$

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Inference rules for implication (function types)

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \supset B \text{ prop}}$$

$$\frac{A \text{ type} \quad B \text{ type}}{A \rightarrow B \text{ type}}$$

- Introduction rule

$$\frac{}{A \text{ true}}^u$$

$$\frac{}{u : A}^u$$

$\vdots$

$\vdots$

$$\frac{B \text{ true}}{A \supset B \text{ true}} \supset I^u$$

$$\frac{M : B}{\lambda u. M : A \supset B} \supset I^u$$

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Inference rules for implication (function types)

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \supset B \text{ prop}} \qquad \frac{A \text{ type} \quad B \text{ type}}{A \rightarrow B \text{ type}}$$

- Introduction rule

$$\frac{}{A \text{ true}}^u \qquad \frac{}{u : A}^u$$
$$\vdots \qquad \vdots$$
$$\frac{B \text{ true}}{A \supset B \text{ true}} \supset I^u \qquad \frac{M : B}{\lambda u. M : A \supset B} \supset I^u$$

- Elimination rule

$$\frac{A \supset B \text{ true} \quad A \text{ true}}{B \text{ true}} \supset E \qquad \frac{M : A \supset B \quad N : A}{M N : B} \supset E$$

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Example

- (i) A proof that $A \supset A$ true.
- (ii) A proof that $\lambda u.u : A \supset A$.

$$\frac{\text{true}}{A \supset A \text{ true}} \supset I^u$$

$$\frac{u : A}{\lambda u.u : A \supset A} \supset I^u$$

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Example

- (i) A proof that $(A \wedge B) \supset (B \wedge A)$ **true**.
- (ii) A proof that $\lambda x. \langle \text{snd } x, \text{fst } x \rangle : (A \wedge B) \supset (B \wedge A)$.

$$\frac{\frac{\frac{}{x} \quad A \wedge B \text{ true}}{B \text{ true}} \wedge E_2 \quad \frac{\frac{}{x} \quad A \wedge B \text{ true}}{A \text{ true}} \wedge E_1}{B \wedge A \text{ true}} \wedge I}{(A \wedge B) \supset (B \wedge A) \text{ true}} \supset I^x$$

$$\frac{\frac{\frac{}{x} \quad x : A \wedge B}{\text{snd } x : B} \wedge E_2 \quad \frac{\frac{}{x} \quad x : A \wedge B}{\text{fst } x : A} \wedge E_1}{\langle \text{snd } x, \text{fst } x \rangle : B \wedge A} \wedge I}{\lambda x. \langle \text{snd } x, \text{fst } x \rangle : (A \wedge B) \supset (B \wedge A)} \supset I^x$$

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Example

- (i) A proof that $A \supset (B \supset A)$ **true**.
- (ii) A proof that $\lambda x. \lambda y. x : A \supset (B \supset A)$.

$$\frac{\frac{\frac{}{A \text{ true}}^u}{B \supset A \text{ true}} \supset I}{A \supset (B \supset A) \text{ true}} \supset I^u$$

$$\frac{\frac{\frac{}{x : A}}{\lambda y. x : B \supset A} \supset I}{\lambda x. \lambda y. x : A \supset (B \supset A)} \supset I^u$$

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Notation

Let Γ be a set of judgements and let $M : A$ be a judgement. The relation $\Gamma \vdash M : A$ means that there is a derivation with conclusion $M : A$ from the set of hypotheses Γ .

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Example

- (i) A proof that $A \wedge B \text{ true} \vdash B \wedge A \text{ true}$.
- (ii) A proof that $M : A \wedge B \vdash \langle \text{snd } M, \text{fst } M \rangle : B \wedge A$.

$$\frac{\frac{A \wedge B \text{ true}}{B \text{ true}} \wedge E_2 \quad \frac{A \wedge B \text{ true}}{A \text{ true}} \wedge E_1}{B \wedge A \text{ true}} \wedge I$$

$$\frac{\frac{M : A \wedge B}{\text{snd } M : B} \wedge E_2 \quad \frac{M : A \wedge B}{\text{fst } M : A} \wedge E_1}{\langle \text{snd } M, \text{fst } M \rangle : B \wedge A} \wedge I$$

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Inference rules for disjunction

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Inference rules for disjunction

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \vee B \text{ prop}}$$

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Inference rules for disjunction

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \vee B \text{ prop}}$$

- Introduction rules

$$\frac{A \text{ true}}{A \vee B \text{ true}} \vee I_1$$

$$\frac{B \text{ true}}{A \vee B \text{ true}} \vee I_2$$

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Inference rules for disjunction

- Formation rule

$$\frac{A \text{ prop} \quad B \text{ prop}}{A \vee B \text{ prop}}$$

- Introduction rules

$$\frac{A \text{ true}}{A \vee B \text{ true}} \vee I_1$$

$$\frac{B \text{ true}}{A \vee B \text{ true}} \vee I_2$$

- Elimination rule

$$\frac{\frac{A \text{ true}}{\vdots}^u \quad \frac{B \text{ true}}{\vdots}^w}{\frac{A \vee B \text{ true} \quad C \text{ true}}{C \text{ true}} \vee E^{u,w}}$$

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Inference rules for truth

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Inference rules for truth

- Formation rule

$$\frac{}{\top \text{ prop}}$$

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Inference rules for truth

- Formation rule

$$\frac{}{\vdash \text{prop}}$$

- Introduction rules

$$\frac{}{\vdash \text{true}} \text{TI}$$

Typed Lambda Calculus (or Proofs as Programs)

Inference rules for truth

- Formation rule

$$\frac{}{\top \text{ prop}}$$

- Introduction rules

$$\frac{}{\top \text{ true}} \top \text{I}$$

- Elimination rule

"We have no information if we know $\top \text{ true}$, so there is no elimination rule." (p. L2.2)

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Inference rules for falsehood

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Inference rules for falsehood

- Formation rule

$$\frac{}{\perp \text{ prop}}$$

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Inference rules for falsehood

- Formation rule

$$\frac{}{\perp \text{ prop}}$$

- Introduction rules

“ $\perp$ is a proposition that should have no proof! Therefore there are no introduction rules.” (p. L2.10)

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Inference rules for falsehood

- Formation rule

$$\frac{}{\perp \text{ prop}}$$

- Introduction rules

“ $\perp$ is a proposition that should have no proof! Therefore there are no introduction rules.” (p. L2.10)

- Elimination rule

$$\frac{\perp \text{ true}}{C \text{ true}} \perp\text{E}$$

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Example

A proof that $\{A \vee B \text{ true}, A \supset C \text{ true}, B \supset C \text{ true}\} \vdash C \text{ true}$.

$$\frac{A \vee B \text{ true} \quad \frac{A \supset C \text{ true} \quad \frac{}{A \text{ true}}^u}{C \text{ true}} \supset E \quad \frac{B \supset C \text{ true} \quad \frac{}{B \text{ true}}^w}{C \text{ true}} \supset E}{C \text{ true}} \vee E^{u,w}$$

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Inference rules for negation

Since we define negation by $\neg A := A \supset \perp$ in the case we talk about **derived** inference rules.

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- Formation rule

$$\frac{A \text{ prop}}{\neg A \text{ prop}}$$

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Inference rules for negation

Since we define negation by $\neg A := A \supset \perp$ in the case we talk about **derived** inference rules.

- Formation rule

$$\frac{A \text{ prop}}{\neg A \text{ prop}}$$

- Introduction rule

$$\frac{\frac{\frac{}{A \text{ true}}^u \quad \vdots \quad \perp \text{ true}}{\neg A \text{ true}}}{\neg I^u}$$

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Inference rules for negation

Since we define negation by $\neg A := A \supset \perp$ in the case we talk about **derived** inference rules.

- Formation rule

$$\frac{A \text{ prop}}{\neg A \text{ prop}}$$

- Introduction rule

$$\frac{}{A \text{ true}}^u$$
$$\vdots$$
$$\frac{\perp \text{ true}}{\neg A \text{ true}} \neg I^u$$

- Elimination rule

$$\frac{\neg A \text{ true} \quad A \text{ true}}{\perp \text{ true}} \neg E$$

References



Frank Pfenning (26th Jan. 2023). Lecture Notes on Proofs as Programs. Material for the course 15-317 Constructive Logic at Carnegie Mellon University, Spring 2023. URL: <https://www.cs.cmu.edu/~fp/courses/15317-s23/> (cit. on pp. 2, 3).



Morten-Heine Sørensen and Paul Urzyczyn (2006). Lectures on the Curry-Howard Isomorphism. Vol. 149. Studies in Logic and the Foundations of Mathematics. Elsevier (cit. on p. 10).